

Yuhan Hu

CONTACT INFORMATION	Edward P. Fitts Department of Industrial and Systems Engineering North Carolina State University – E-mail: yhu27@ncsu.edu , – Phone: +1 (919) 757-7661
RESEARCH INTERESTS	My research interests include statistical machine learning and multimodal AI, with applications in industrial system management and healthcare. I am particularly interested in: – Development of advanced statistical and federated learning algorithms for prognostics – Stochastic process-based reliability analysis with uncertainty and risk considerations – Calibration and evaluation for reliable multimodal LLMs
EDUCATION	Ph.D., Industrial and Systems Engineering , North Carolina State University Expected 2026 – Dissertation title: <i>Statistical and machine learning approaches for system reliability modeling and prognostics</i>. Advisors: Profs. Xiaolei Fang, Shu-Cherng Fang, and Mengmeng Zhu M.S., Statistics; M.S., Industrial and Systems Engineering; M.S., Textile Engineering , North Carolina State University 2021-2026 B.S., Fashion Design and Engineering , Zhejiang Sci-Tech University 2020
RESEARCH EXPERIENCE	North Carolina State University – Department of Industrial and Systems Engineering <i>Research Assistant</i> Aug. 2021-May 2025 – <i>Personalized federated failure-time prediction</i> • Developed a privacy-preserving failure-time prediction model using log-location-scale regression to address data heterogeneity. – <i>Federated data fusion-based prognostic model</i> • Designed an end-to-end privacy-preserving prognostic pipeline for high-dimensional, outlier-contaminated data that prioritizes user privacy while maintaining model accuracy. – <i>Reliability modeling with damage resistance</i> • Proposed a two-stage reliability model that captures the interactions among degradation, random shocks, and damage resistance. – <i>Human-machine system reliability modeling</i> • Proposed a human-machine interaction system reliability model that integrates situation awareness, machine degradation, and external shocks. Duke University – Department of Computer Science <i>Research Team Member</i> Jun. 2025-Present – <i>Vision-Dependent Mathematical Reasoning Benchmark</i> • Developed a diagnostic decoupling framework on mathematical problems with visual context to disentangle visual perception errors from reasoning failures. UNC Greensboro – Department of Computer Science <i>Research Team Member</i> Aug. 2025-Present – <i>Modality-Aware Calibration for Multimodal LLMs</i> • Proposed a calibration framework for multimodal LLMs, which aligns modality importance with model reliance to mitigate unreliable inference.

JOURNAL PAPERS	(J1) Hu, Y., Zhu, M., & Lin, H. (2025). A nonlinear Wiener process degradation model with damage resistance for reliability analysis. <i>Annals of Operations Research</i>, 1-24. (J2) Hu, Y., Wang, R., Zhu, M., & Chen, K. B. (2023). Modeling human-machine interaction system reliability with multiple dependent degradation processes and situation awareness. <i>International Journal of Reliability, Quality and Safety Engineering</i>, 30(04), 2350014. (J3) Hu, Y., & Fang, X. (2026). A federated learning approach for personalized prognostics in the presence of user-level heterogeneity. Submitted to <i>IISE Transactions</i>.
BOOK CHAPTER	(B1) Hu, Y., & Zhu, M. (2022). System reliability models with random shocks and uncertainty: A state-of-the-art review. <i>Predictive Analytics in System Reliability</i>, Springer.
CONFERENCE PAPERS	(C1) Hu, Y., & Zhu, M. (2022). Reliability modeling for shock-degradation-dependent process considering damage resistance. <i>Proceedings of the 27th ISSAT International Conference on Reliability and Quality in Design</i>. (C2) Sultana, A., Hu, Y., & Sun, Y. (2026). Modality importance and reliance-aware calibration for robust LLM inference. Submitted to <i>AAAI</i>.
PAPERS IN PREPARATION	(P1) Hu, Y., & Fang, X. Towards failure-mode-aware federated prognostics: Selective collaboration for failure time prediction. (P2) Hu, Y., Wang, J., Vartak, R., & Dhingra, B. Enhancing mathematical reasoning of LLMs in visual contexts.
TEACHING EXPERIENCE	Instructor Aug. 2026 – Designed course materials and taught the basics of linear algebra to incoming ISE and OR PhD students to prepare them for advanced doctoral courses. Teaching Assistant – ISE/OR/MA 766: Network Flows, <i>Graduate level</i> Fall 2026 – ISE 443: Quality Design and Control, <i>Undergraduate level</i> Spring 2026 – ISE 437: Data Analytics for Industrial Engineering, <i>Undergraduate level</i> Fall 2025 – ISE 537: Statistical Models for Systems Analytics in Industrial Engineering, <i>Graduate level</i> Spring 2023, Fall 2024 – ISE 311: Engineering Economics Analysis, <i>Undergraduate level</i> Fall 2023, Spring 2024 • Led in-class discussions and course project presentations, and graded homework and exams. • Held office hours to answer students' questions and address their learning difficulties. Teaching Associate 2021-2022 – Mentored undergraduates to perform battery degradation instrumentation and tests – Managed database and coordinated IT services in the College of Textiles Advanced Teaching Certificate Summer 2025 – Completed program in pedagogy, mentoring, and teaching demonstrations.
HONORS AND AWARDS	– INFORMS Student Chapter Annual Award 2025 – NC State University Provost's Fellowship 2022 – Zhejiang Sci-Tech University Undergraduate Scholarship Award 2017, 2018, 2019

CONFERENCE PRESENTATIONS AND TALKS	1. “Towards failure-mode-aware federated prognostics: selective collaboration for failure time prediction.” 2025 INFORMS Annual Meeting, Atlanta, GA	Oct. 2025
	2. “A personalized federated data analytical model for RUL prediction.” 2024 INFORMS Annual Meeting, Seattle, WA	Oct. 2024
	3. “A federated learning approach for personalized prognostics in the presence of user-level heterogeneity.” ISE Ph.D. Student Seminar, Raleigh, NC	Sept. 2024
	4. “A nonlinear Wiener process degradation model with damage resistance for reliability analysis” (poster). 2023 INFORMS Annual Meeting, Phoenix, AZ	Oct. 2023
	5. “Reliability modeling for shock-degradation-dependent process considering damage resistance.” 27th ISSAT Int’l Conf. on Reliability and Quality in Design, Virtual	Aug. 2022
	6. “Human-machine interaction system reliability model with multiple dependent degradation processes.” 2021 INFORMS Annual Meeting, Virtual	Oct. 2021
SERVICES	President, INFORMS NCSU Student Chapter , NC State University	Aug. 2024-May 2025
	– Host seminars, job-search talks, and programming workshops	
	– Received 2 nd place in the INFORMS PhD student chapter video contest	
	Secretary, INFORMS NCSU Student Chapter , NC State University	Aug. 2023-May 2024
	– Coordinated with relevant teams and departments, ensuring effective updates to the president	
	Reviewer for <i>Reliability Engineering & System Safety</i> , <i>IEEE Transactions on Reliability</i> , <i>The Engineering Economist</i>	
REFERENCES	Dr. Shu-Cherng Fang Walter Clark Chair Professor and University Alumni Distinguished Graduate Professor Department of Industrial and Systems Engineering, North Carolina State University Phone: 919-515-2192 E-mail: fang@ncsu.edu Dr. Xiaolei Fang Associate Professor Department of Industrial and Systems Engineering, North Carolina State University Phone: 404-542-8524 E-mail: xfang8@ncsu.edu Dr. Russell King Dopaco Distinguished Professor Department of Industrial and Systems Engineering, North Carolina State University Phone: 919-515-5186 E-mail: king@ncsu.edu	